A Research Proposal

"Market Making under Cross-Asset Hawkes Order Flow: Closed-Form Approximations, Exact Solutions and Reinforcement Learning in One Study"

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Abstract

Market makers quote bid and ask prices for many assets at once, and the orders they serve do not arrive at a constant rate. Orders come in bursts, and a burst in one asset may be followed by orders in related assets. Self-exciting and mutually exciting point processes, known as Hawkes processes, describe this clustering well. The closed-form theory of market making, from the single-asset model of Guéant, Lehalle and Fernandez-Tapia (2013) to the multi-asset model of Bergault, Evangelista, Guéant and Vieira (2021), assumes instead that orders arrive as independent Poisson processes. This research aims to extend both models to Hawkes order flow, first for one asset and then for several assets with excitation across assets, and to compare three ways of solving the resulting control problem within one study: closed-form approximations, exact numerical solutions and reinforcement learning. A systematic literature search carried out for this proposal screened 75 papers and found no study that gives a closed-form approximation for multi-asset market making with cross-asset Hawkes flow, and no published study that compares the three methods on the same Hawkes problem. The study will derive the approximations, compute exact reference solutions, train learning agents on the true event simulator and score every policy on the same simulated order flow, both for parameter sets taken from the literature and for Hawkes models fitted to trade and order book data from cryptocurrency and equity index futures. The expected outcome is a map of where each method stays close to the optimum and where it does not, together with code and run records that make every reported number reproducible.

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List of Main Symbols

d	Number of assets quoted by the market maker
S_t, σ, Σ	Reference prices, volatility, and the covariance matrix of price changes
q_t, Q, X_t	Inventory (vector of positions), inventory bound, and cash
δ^b, δ^a	Distances of the bid and ask quotes from the reference price
Λ, k	Intensity scale and fill decay of the 2013 model, whose fill intensity is $\Lambda e^{-k\delta}$
N_t^j	Number of market orders of type j (one asset and one side) up to time t
λ_t, λ^{+j}	Vector of order intensities, and its value just after an order of type j
μ, $\bar{\lambda}$	Baseline intensities and stationary mean intensities
K, $\rho(K)$	Branching matrix (K_{jl} is the mean number of type j orders triggered by one type l order) and its spectral radius; the flow is stationary when $\rho(K) < 1$
β, D	Decay rates of the exponential kernels, and $D = \mathrm{diag}(\beta)$
$f(\delta)$	Probability that an arriving order fills a quote at distance δ
z_i	Trade size in asset i
$H^{i,s}$, $h_2^{i,s}$	Hamiltonian function of asset i on side s, and its second derivative at zero
p^j	Shadow price of inventory used to set the quote met by an order of type j
γ	Risk aversion of the market maker
$m_t(u)$	Forecast, made at time t, of the intensity u time units ahead
$\theta(t, q, \lambda)$	Value function after the cash and the marked-to-market inventory are factored out
A, B, C	Coefficients of the quadratic approximation $\theta \approx -q^\top A q - q^\top B - C$
T	Time horizon

1. Introduction

Market makers provide liquidity to other traders. They post prices at which they are ready to buy (the bid) and to sell (the ask), earn the difference between the two when both sides trade, and carry the inventory that builds up when they do not. The mathematical study of this problem goes back to Ho and Stoll (1981). In its modern form, introduced by Avellaneda and Stoikov (2008), the reference price follows a Brownian motion and market orders reach a quote at distance δ from the reference price as a Poisson process whose intensity falls exponentially with δ. The market maker chooses the two quotes to balance the spread earned against the risk of the inventory.

Guéant, Lehalle and Fernandez-Tapia (2013) gave this model a rigorous solution. With inventory bounds, a change of variables turns the Hamilton-Jacobi-Bellman equation into a system of linear ordinary differential equations, which gives the optimal quotes through a matrix exponential and leads to closed-form approximations of the quotes. Bergault, Evangelista, Guéant and Vieira (2021) extended the closed-form approach to many assets. They replaced the Hamiltonian functions of the problem by quadratic functions, which reduces the value function to a quadratic form in the inventory whose coefficients solve a matrix Riccati equation. This gives quotes in closed form for portfolios of assets where numerical grids are out of reach, and the authors also present these formulas as starting points for learning methods.

Both models assume that orders arrive as Poisson processes, so that the number of orders in one period says nothing about the next. Real order flow does not behave this way. Orders come in bursts, and one order makes further orders more likely for a short time. Hawkes processes (Hawkes, 1971) capture this effect and are now a standard model of order flow (Bacry, Mastromatteo and Muzy, 2015). In a preliminary analysis for this study, a bivariate Hawkes model fitted to bursts of market orders (one event per side and 100 millisecond bin) on five Binance futures contracts beat a Poisson model on every contract. The branching ratios were between 0.44 and 0.93, which means that between 44% and 93% of the bursts were triggered by earlier bursts.

Market makers also quote several assets at once, and the inventory risk they carry is a portfolio risk. When the flow of one asset excites the flow of another, for example when a wave of sell orders hits two cryptocurrency futures traded by largely the same participants, the market maker builds a joint position faster than independent flows would allow. The size of this cross-asset excitation is an empirical question, and its effect on optimal quotes has not been worked out in closed form.

A systematic literature search carried out for this proposal found 21 papers on Hawkes order flow in market making: 19 market making studies and two papers on the theory of the Markov lift that such

studies need. Only two of them keep a closed form: the first-order expansion of Cartea, Jaimungal and Ricci (2014) and the Volterra-Riccati approximation of Barzykin (2026b). Both treat a single asset. The few multi-asset Hawkes studies use learned policies only, and the one with explicit cross-asset excitation (Acero, Plank and Zhang, 2026) compares its reinforcement learning agents only with other neural networks. No published study compares a closed-form approximation, an exact solution and a reinforcement learning agent on the same Hawkes problem.

Therefore, this study intends to fill that gap in two stages. The first stage takes the single-asset model of 2013 and solves it under Hawkes order flow. The second stage takes the multi-asset model of 2021 and solves it under Hawkes order flow with excitation across assets. In both stages the problem will be solved in three ways: by closed-form approximation, by an exact numerical method and by reinforcement learning. The three will be scored on the same simulated order flow. The results are meant to answer three questions for clustered order flow: whether a closed form exists, whether deep learning can solve the problem, and whether deep learning adds value over the closed forms.

1.1 Rationale of the study

A closed-form quoting rule is only as good as the model of order flow behind it. The motive for this research is developed from the following rationale:

To Address the Knowledge Gap: The Hawkes literature on market making is split into closed-form work on one asset, numerical solutions of lifted equations on one asset, and reinforcement learning studies without an exact benchmark. The multi-asset closed form of 2021 has not been extended to Hawkes flow, and the single-asset approximation of Barzykin (2026b) has been checked against an exact solution only for one asset and one exponential memory variable. This study will address this gap by deriving the multi-asset approximation and testing it against exact solutions.

Its Relevance to Market Practice: Dealers in bonds, foreign exchange and cryptocurrencies quote portfolios, not single assets, and their order flow is clustered. A closed-form rule that reacts to a burst of orders in one asset by adjusting the quotes of a correlated asset would be directly usable, because it costs one matrix Riccati equation per update instead of the solution of a high-dimensional equation. Grids, which give the exact answer for one asset, become impractical beyond two or three assets.

Methodological Relevance: Each solution method, used alone, can mislead. A closed-form approximation hides its own error. In a reproduction of the exponential benchmark of Barzykin (2026b), carried out as preliminary work for this study, the exact values matched the paper, but

the state-feedback policy lost about 0.6% against the exact optimum in the directional regime, against 0.17% reported in the paper. Most of this loss came from the quadratic expansion of the Hamiltonian: with the exact Hamiltonian the same state-feedback construction lost 0.25%, and the version with the pre-order forecast lost 0.09%. A second, independent implementation confirmed these numbers. A reinforcement learning result without an exact benchmark can also be far from the optimum. In a simulation of the single-asset environment of Acero, Plank and Zhang (2026), also made as preliminary work, a constant quote earned more than the agents reported in that paper. This is a flag rather than a refutation, since their code is not public. An exact method, finally, does not scale. Comparing the three on the same order flow separates the error of each.

At the end, this research will give quoting rules for clustered, multi-asset order flow, together with regret tables that show where each rule stays close to the optimum and where it does not.

1.2 Basic Concepts and Definitions

This section defines the terms that are used throughout the proposal.

1) Market making and optimal quotes

A market maker continuously posts a bid and an ask price around a reference price S_t. The distances δ^b and δ^a of the two quotes from S_t are the controls. A smaller distance is filled more often but earns less per trade. The optimal quotes maximize an objective that rewards the spread earned and penalizes the inventory q_t, either through a utility function (the 2013 model) or through a running penalty proportional to the variance of the inventory value (the risk-adjusted objective used in the 2021 model).

2) Hawkes process

A Hawkes process is a point process whose intensity rises after each event and then decays back toward a baseline. With an exponential kernel, the intensity of orders of type j is

$$\lambda_t^j = \mu_j + \sum_l K_{jl} \int_0^t \beta_j e^{-\beta_j (t-s)} \, \mathrm{d}N_s^l,$$

where μ_j is the baseline, β_j the decay rate and K_{jl} the mean number of type j orders triggered by one type l order. The flow is stationary when the spectral radius $\rho(K)$ is below one. For a single asset with symmetric excitation, $\rho(K)$ is the branching ratio, the share of orders triggered by earlier orders.

3) Speed of the memory

After a burst, the forecast of the total activity relaxes toward its mean at the rate $\beta(1-\rho(K))$, which is slower than the decay rate β of the kernel itself. In this proposal the memory is called slow when this relaxation rate is small compared with the mean arrival rate of orders, and fast otherwise. In the preliminary Binance fits, the forecast half-life was about 140 seconds for BTCUSDT (slow memory) and about 1.2 seconds for MKRUSDT (fast memory, about two seconds between bursts on each side).

4) Cross-asset excitation

Cross-asset excitation means that $K_{jl} > 0$ for order types j and l that belong to different assets, so that orders in one asset raise the intensity of orders in another. It is a property of the flow, separate from the correlation of the price changes.

5) Closed-form approximation

A closed-form approximation replaces the exact value function by one that can be written with explicit formulas or with the solution of a small system of ordinary differential equations. In this proposal it means the linear system of the 2013 model and the matrix Riccati equation of the 2021 model, driven by a forecast of the Hawkes intensities.

6) Exact solution

With exponential kernels the intensities form a Markov state, and the problem becomes a Hamilton-Jacobi-Bellman equation in the inventory and the intensities. A numerical solution of this equation on a grid, with the discretization error reduced until it is negligible, is called the exact solution in this proposal.

7) Reinforcement learning

Reinforcement learning finds a quoting policy by interacting with a simulator of the market, without solving the equation. In this study the agents are trained on an exact simulator of the Hawkes flow, so that their errors come from learning and not from the simulator.

8) Regret

The regret of a policy is its loss relative to the exact optimum, in percent of the optimal value. Under the utility objective of the 2013 model it is measured on the certainty equivalent $-\frac{1}{\gamma}\ln(-u)$, which is expressed in money. Where no exact solution is available, the loss is measured against the best policy found. Both are estimated by Monte Carlo simulation, with all policies facing the same simulated orders and prices.

1.3 Statement of the problem

Despite the evidence that order flow is clustered, the closed-form theory of market making still rests on Poisson arrivals, and its extension to clustered flow is incomplete. Questions remain:

- ➢ How accurate is the linear structure of the 2013 solution, which survives under Hawkes order flow only in approximate form?
- ➢ How should the matrix Riccati system of the 2021 model change when the order flow of one asset excites the flow of another, and what does a market maker lose by ignoring this excitation?
- ➢ How far are such closed-form approximations from the exact optimum, and can a reinforcement learning agent close the remaining gap?

Without a study of this kind, a market maker cannot know whether a closed-form rule is accurate enough under clustered flow, and a researcher cannot know whether a gain reported for a learning agent is real. This study will address the gap by solving the same Hawkes problems in three ways and comparing the answers on the same order flow.

2. Research Aim and Objectives

2.1 Aim

- ❖ To extend the closed-form market making models of 2013 (one asset) and 2021 (several assets) to Hawkes order flow with excitation across assets, and to measure, within one study, how closed-form approximations, exact solutions and reinforcement learning compare on the same problem.

2.2 Specific Objectives

- ✓ To formulate the single-asset model of 2013 with Hawkes order flow and to derive closed-form approximations of its optimal quotes.
- ✓ To compute the exact solution of the single-asset problem as a benchmark for these approximations.
- ✓ To formulate the multi-asset model of 2021 with a multivariate Hawkes flow that includes cross-asset excitation, and to derive a matrix Riccati approximation driven by the forecast of all intensities.
- ✓ To build exact or near-exact reference solutions for two assets.
- ✓ To train reinforcement learning agents in the same environments, starting from the closed-form policies.
- ✓ To compare all policies on common simulated order flow, for parameter sets from the literature and for Hawkes models fitted to market data from cryptocurrency and equity index futures.

2.3 Research Questions

The central question for this research that the researcher intends to answer is

- ❖ Can the closed-form approach of the 2013 and 2021 market making models be extended to Hawkes order flow with excitation across assets, and how do such closed forms compare with the exact optimum and with reinforcement learning? This central question can be split into the following detailed questions:

- ✓ How accurate is the 2013 linear system when the constant intensity is replaced by the forecast of a Hawkes intensity?

- ✓ How does cross-asset excitation enter the matrix Riccati system of the 2021 model, and how much value does a market maker lose by ignoring it?
- ✓ Which part of the error of a closed-form policy comes from freezing the intensity forecast, and which part from the quadratic expansion of the Hamiltonian?
- ✓ Under which conditions (speed of the memory, branching ratio, strength of the cross-excitation, price correlation, risk aversion) does a reinforcement learning agent beat the closed forms?
- ✓ Do the answers hold for Hawkes models fitted to real order flow?

2.4 Hypotheses

H1: For a single asset with slow memory, the 2013 linear system driven by the Hawkes intensity forecast stays within 0.1% of the certainty equivalent of the exact optimum, and its regret grows as the memory becomes faster.

H2: For several assets, a policy that uses the correct mean intensities but sets the cross-asset blocks of K to zero in its forecast loses a share of the risk-adjusted revenue that grows with the strength of the cross-asset excitation, and the loss is largest when the excitation and the price correlation push the inventories in the same direction.

H3: Where the Riccati-based policies lose most, the larger part of their error comes from the quadratic expansion of the Hamiltonian and not from freezing the forecast, so that the same forecast combined with the exact Hamiltonian stays close to the optimum.

H4: A reinforcement learning agent trained on the true event simulator and started from the closed-form policy improves on the closed forms where they are weak (fast memory, high risk aversion), and stays behind them where the memory is slow.

3. Literature Review

3.1 Theoretical Basis

The theory of dealer pricing under inventory risk starts with Ho and Stoll (1981), who showed that a dealer facing uncertain order arrivals and price changes should shade both quotes against the inventory. Avellaneda and Stoikov (2008) gave the model its current form: a Brownian reference price, Poisson arrivals of market orders with intensity $\Lambda e^{-k\delta}$ at distance δ, and a market maker with exponential utility over a finite horizon. They solved the resulting Hamilton-Jacobi-Bellman equation approximately. (The original papers write the intensity scale as A; this proposal writes Λ and keeps A for the Riccati matrix below.)

Guéant, Lehalle and Fernandez-Tapia (2013), a paper circulated as a preprint from 2011, solved the same model under inventory bounds. They showed that the change of variables $u = -\exp(-\gamma(x + qs))\, v_q(t)^{-\gamma/k}$ reduces the Hamilton-Jacobi-Bellman equation to the linear system $v_q' = \frac{k\gamma\sigma^2}{2} q^2 v_q - \eta(v_{q-1} + v_{q+1})$, with $\eta = \Lambda(1 + \gamma/k)^{-(1+k/\gamma)}$, which is solved by a matrix exponential. They also proved a verification theorem, studied the asymptotic regime through a spectral problem and derived closed-form approximations of the quotes. Guéant (2017) extended the characterization to general intensity functions and to several assets, and Cartea, Jaimungal and Penalva (2015) use a risk-adjusted objective, in which a running penalty on the inventory replaces the utility function.

Bergault, Evangelista, Guéant and Vieira (2021) addressed the problem of many assets, where grids fail. Their idea is to replace each Hamiltonian function by a quadratic function of the shadow price. The approximate value function is then a quadratic form $-q^\top A(t) q - q^\top B(t) - C(t)$ in the inventory, and A solves a matrix Riccati equation driven by the covariance matrix of prices and by the order intensities. The quotes follow in closed form, both over a finite horizon and in the long-horizon limit, and the authors suggest using these formulas as heuristic evaluation functions and as starting points for reinforcement learning.

Hawkes (1971) introduced self-exciting and mutually exciting point processes. In finance they are used to model the clustering of trades, quotes and price changes (Bacry, Mastromatteo and Muzy, 2015), and estimates on order book data often show high branching ratios and slowly decaying kernels (Bacry, Jaisson and Muzy, 2016). An exponential kernel makes the intensities a Markov state, and a sum of exponentials adds one state variable per term. General kernels can be approximated by such sums or by other finite expansions, with convergence of the controlled problem (Khabou and Talbi, 2025; Martin and Tse, 2026). This Markov lift is what makes an exact solution of the

Hawkes market making problem possible.

On the learning side, Guéant and Manziuk (2019) solved multi-asset market making with an actor-critic method for a large number of corporate bonds, under Poisson flow. Their method is model-based, in the sense that it uses the known dynamics of the problem when it updates the value function and the policy. It is the first of the two reinforcement learning methods used in this study.

3.2 Related Studies

3.2.1 Hawkes Order Flow in Closed-Form Market Making Models

Cartea, Jaimungal and Ricci (2014) were the first to put mutually exciting market orders into a market making model. In their model, influential market orders excite both sides and move the depth of the book, and the market maker maximizes expected terminal wealth with a penalty on the inventory. They obtained quotes that are linear in the difference between the buy and sell intensities through a first-order expansion in the inventory penalty, and they tested the strategy by simulation. The model has one asset and no comparison with an exact solution.

Barzykin (2026b) placed Hawkes requests for quotes inside the closed-form framework of 2021. He fitted marked Hawkes processes with three exponential components to three years of foreign exchange requests and found branching ratios between 0.86 and 0.90. He then replaced the random future intensity by its conditional forecast, which solves a Volterra equation, and solved the Riccati system of 2021 along this forecast. He added a covariance correction and a state-feedback version that updates the forecast after each request. Against the exact solution of the lifted equation for a single exponential memory, the state-feedback policy lost between 0.08% and 0.17%, while a Poisson policy lost between 6% and 27%. He also stated that a Markov lift combined with the quadratic approximation gives transport-jump equations in the memory variables rather than a small Riccati system. All his experiments use one asset, and the paper contains no learning method. The preliminary reproduction of this benchmark matched the exact values and the Poisson regrets, but not the 0.17% of the state-feedback policy. It also showed that the Poisson comparator uses the baseline intensity, which is far below the mean intensity, so much of its large regret reflects a weak comparator.

Bergault, Bertucci, Bouba, Guéant and Guilbert (2024) showed, for automated market makers, why the quadratic reduction fails when the trades themselves excite the flow: the inventory and the intensity then jump together and the ansatz does not close. They used Markov-modulated Poisson flow instead. In two papers, Barzykin (2026a, 2026c) let the dealer's own rejections or wins feed

back into the request flow through slow scores, and kept a Riccati approximation by freezing the slow state, with a check against the exact solution in one case.

3.2.2 Exact and Numerical Solutions under Hawkes Order Flow

Jusselin (2021) solved market making with persistent order flow, where the market maker is the only liquidity provider and his own trades excite the flow. With a completely monotone kernel approximated by a sum of exponentials, he characterized the value function through a lifted equation and solved it by finite differences, comparing Hawkes-aware and Poisson-based controls. Martin and Tse (2026) treated general kernels by a Markov lift based on Mercer's expansion, proved convergence of the lifted value functions, and solved the lifted equation with a physics-informed neural network. Their model also lets the fill decay parameter be excited by market orders. It has one asset and no buy and sell cross-excitation, and the neural solution is not checked against a grid solution.

Choi, Jang, Lee and Zheng (2021) used Hawkes market orders with self, mutual and synchronising excitation, fitted by maximum likelihood to trades of six large United States stocks, and solved the equation with the deep Galerkin method. Tarditi (2026), in a master's thesis read at the level of its abstract, calibrated a bivariate Hawkes process on order book data and solved the lifted equation on a grid. Jain, Firoozye, Kochems and Treleaven (2025) built a Hawkes order book with twelve event types and found that the deep Galerkin method did not converge, while a policy gradient agent did. None of these studies derives a closed-form approximation, and all of them treat a single asset.

3.2.3 Reinforcement Learning under Hawkes Order Flow

Reinforcement learning is the largest group in the Hawkes literature, with twelve of the 21 papers found. Acero, Plank and Zhang (2026) proposed networks with learnable exponential filters, trained by proximal policy optimization (Schulman, Wolski, Dhariwal, Radford and Klimov, 2017), and proved a regret bound for policies built on a truncated kernel. They ran single-asset and multi-asset experiments, the latter with cross-asset excitation, but every baseline is another neural network, so the distance of the agents from the optimum is unknown. Gašperov and Kostanjčar (2022) trained a soft actor-critic agent in an eight-variate Hawkes order book and compared it with heuristic rules. Yang and Xu (2026) let the market maker's own fills excite the intensity and trained an adversarial agent. Lai, Gao and Li (2023) fitted power-law Hawkes processes to option market orders, with strong self-excitation and insignificant cross-excitation between buys and sells, and trained a neural policy that beat a Poisson-trained one. Kumar (2024) drove a multi-asset agent-based order book with a neural Hawkes model.

The two remaining papers study the Markov lift itself. Besides Khabou and Talbi (2025), Bielecki, Mastrolia and Yan (2026) developed, according to their abstract, continuous-time reinforcement learning for controlled Hawkes jump-diffusions; in the researcher's reading, this tool fits the lifted problem of this study.

3.2.4 Multi-Asset Closed Forms beyond Poisson Order Flow

Several papers keep the Riccati structure of 2021 when the Poisson assumption is relaxed in other ways. Bergault and Guéant (2023) let the request intensities follow a Markov-modulated Poisson process with four observed regimes and obtained one Riccati equation per regime, coupled through the transition rates, for a single asset. Barzykin, Bergault and Guéant (2023) solved multi-currency market making with client tiers, pricing ladders and hedging through a matrix Riccati equation and checked it against a grid for two currencies. Barzykin and Ciceri (2026) added a hit-ratio target to bond market making on a size ladder. Boyce, Herdegen and Sánchez-Betancourt (2025) studied market making against exogenous competition and compared a closed form, a numerical solution and a policy gradient agent in one study, under Poisson flow. Their closed form was within 1% of the numerical solution, and the agent was clearly behind both. Rosenbaum and Zhang (2022) obtained asymptotic closed forms for multi-asset market making under a rough volatility model, where the Volterra structure lies in the prices and not in the order flow.

3.3 Gaps in Literature

The systematic search carried out for this proposal gives the following picture. Of the 75 papers screened, 57 were kept. Twenty-one of these concern Hawkes order flow: two market making studies keep a closed form, five solve a lifted equation numerically, twelve use reinforcement learning or another learned policy, and two study the Markov lift itself without a market making application. For the other execution models the picture is the opposite, since at least 20 of 36 papers keep a Riccati equation or another closed form.

Two combinations were not found by this search. First, no paper extends the multi-asset closed form of 2021 to Hawkes order flow with cross-asset excitation. Every Hawkes closed form or approximation treats a single asset, Martin and Tse (2026) leave even the buy and sell cross-excitation of one asset to future work, and the multi-asset Hawkes studies (Lai, Gao and Li, 2023; Kumar, 2024; Acero, Plank and Zhang, 2026) use learned policies alone. Second, no published paper compares a closed-form approximation, an exact solution and a reinforcement learning agent on the same Hawkes problem. Each study has at most two of the three, and the one study that has all three

(Boyce, Herdegen and Sánchez-Betancourt, 2025) uses Poisson flow.

These gaps matter for the reasons given in Section 1.1. A closed form without an exact benchmark can hide an error of the same size as the effect it models, and a learning agent without one can look good against weak baselines. The preliminary work for this study points the same way. Under Poisson flow, the closed form of 2021 was within 0.05% of the optimum for single bonds but lost 0.58% and 1.39% in two harder cases, and an actor-critic agent beat it for a pair of bonds. Under Hawkes flow fitted to Binance data, for one asset and the risk-adjusted objective, ignoring clustering cost between 0.3% and 0.6% of the risk-adjusted revenue (0.9% at the higher risk aversion). The Hawkes-aware closed form was the best policy when the memory was slow, and a reinforcement learning agent trained on the true simulator was the best policy short of the exact grid solution when the memory was fast.

Therefore, this study will contribute by deriving the closed-form approximations of the 2013 and 2021 models under Hawkes flow, including cross-asset excitation, and by comparing them with exact solutions and with reinforcement learning on the same order flow. A search cannot rule out theses or unindexed drafts, so the gaps are stated as not found by this search rather than as nonexistent.

4. Methodology

4.1 Research Design

A two-stage quantitative design will be used. The first stage solves the single-asset model of 2013 under Hawkes order flow and the second stage solves the multi-asset model of 2021 under Hawkes order flow with cross-asset excitation. In each stage the problem is solved in three ways, and all solutions are scored by the same evaluation protocol. Table 1 summarizes the design.

	Closed-form approximation	**Exact or near-exact solution**	**Reinforcement learning**
Stage 1: one asset, 2013 model	Linear system of 2013 driven by the intensity forecast; single-asset Volterra-Riccati policy	Lifted Hamilton-Jacobi-Bellman equation on a grid in $(q, \lambda^b, \lambda^a)$	Actor-critic and policy gradient agents trained on the event simulator
Stage 2: two or more assets, 2021 model	Matrix Volterra-Riccati policy with the cross-asset forecast; for two assets, the same forecast with the exact Hamiltonian	Grid for reduced memory models; for the full two-asset model, a neural solver if it passes validation on them	The same agents with the joint state
Evaluation: Monte Carlo simulation of the true Hawkes model, all policies on the same orders and prices; regret against the exact solution where one exists, otherwise loss against the best policy found.			

Table 1: Design of the study.

Stage 1, the 2013 model with Hawkes order flow: The reference price follows $\mathrm{d}S_t = \sigma\,\mathrm{d}W_t$. Sell market orders, which hit the bid, and buy market orders, which lift the ask, form a bivariate Hawkes process with intensities

$$\lambda_t^b = \mu + \int_0^t \beta e^{-\beta(t-s)}\left(\alpha_s\,\mathrm{d}N_s^b + \alpha_c\,\mathrm{d}N_s^a\right), \qquad \lambda_t^a = \mu + \int_0^t \beta e^{-\beta(t-s)}\left(\alpha_c\,\mathrm{d}N_s^b + \alpha_s\,\mathrm{d}N_s^a\right), \quad (1)$$

where α_s measures self-excitation and α_c excitation across the two sides. An arriving order fills the quote at distance δ with probability $e^{-k\delta}$. The fill intensity of the bid is therefore $\lambda_t^b e^{-k\delta_t^b}$, which is the intensity $\Lambda e^{-k\delta}$ of the 2013 model with Λ replaced by λ_t^b. Every market order excites the flow, whether it fills the market maker or not. The market maker maximizes $\mathbb{E}[-\exp(-\gamma(X_T + q_T S_T))]$

under the bound $|q_t| \leq Q$, as in the 2013 paper. A single-asset comparison of closed forms, the exact solution and an actor-critic agent under the risk-adjusted objective was made as preliminary work; Stage 1 extends it to the utility objective and the linear system of 2013, and adds the Volterra-Riccati policy of Barzykin (2026b).

The exact problem has the state $(t, q, \lambda^b, \lambda^a)$. No exponential change of variables linearizes the Hawkes equation, because the value of an order that does not fill the market maker enters non-linearly. The linear structure survives, however, along a deterministic intensity path: the change of variables of 2013 still applies when the intensity is a known function of time $\Lambda(t)$. Given the information at time t, the forecast of the intensity vector is

$$m_t(u) = \bar{\lambda} + e^{-\beta(I-K)u}\big(\lambda_t - \bar{\lambda}\big), \qquad K = \begin{pmatrix} \alpha_s & \alpha_c \\ \alpha_c & \alpha_s \end{pmatrix}, \qquad \bar{\lambda} = \frac{\mu}{1-\alpha_s-\alpha_c}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{2}$$

so the total activity relaxes at the rate $\beta(1-\alpha_s-\alpha_c)$ and the imbalance between the two sides at the rate $\beta(1-\alpha_s+\alpha_c)$. With the intensity frozen at this forecast, the change of variables of 2013 gives the linear system with time-varying coefficients

$$\frac{\mathrm{d}v_q}{\mathrm{d}\tau} = \frac{k\gamma\sigma^2}{2}q^2 v_q - c\, m_t^b(\tau - t)\, v_{q+1} - c\, m_t^a(\tau - t)\, v_{q-1}, \qquad v_q(T) = 1, \tag{3}$$

for $\tau \in [t, T]$, where $c = (1+\gamma/k)^{-(1+k/\gamma)}$, with the boundary rows of the 2013 paper at $q = \pm Q$. The quotes then follow from the formulas of 2013,

$$\delta^b(t,q) = \frac{1}{k}\ln\frac{v_q(t)}{v_{q+1}(t)} + \frac{1}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big), \qquad \delta^a(t,q) = \frac{1}{k}\ln\frac{v_q(t)}{v_{q-1}(t)} + \frac{1}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big). \tag{4}$$

A first numerical check, made while preparing this proposal, confirmed equation (3): with deterministic bid and ask intensities that vary in time, the linear system and a direct solution of the nonlinear equation agree to the tolerance of the solver. Between orders the forecast moves deterministically, so the system needs to be solved again only when an order arrives. A state-feedback version evaluates it at the forecast just after the arriving order, because the quote that the order meets should be set against the value after the excitation it causes.

Because the linear system of 2013 is exact for a deterministic intensity path, its only error under Hawkes flow is the freezing of the forecast. The quadratic Volterra-Riccati policy of Barzykin (2026b) adds the error of the expansion of the Hamiltonian. Comparing the two on the same flow therefore separates the two sources of error. The study will also compare them with the 2013 quotes at the mean intensity, with the 2013 quotes at the current intensity, and with the exact solution of the lifted equation. The exact solution will be computed on a grid in $(q, \lambda^b, \lambda^a)$ by a semi-Lagrangian

scheme, which transports the solution along the decay of the intensities and applies the jumps at the post-order intensities. The same scheme, with a single memory variable, reproduced the exact values of the benchmark of Barzykin (2026b) in the preliminary work. Convergence in the grid steps and the time step will be reported for every exact value.

Stage 2, the 2021 model with cross-asset Hawkes order flow: There are d assets with reference prices $S_t \in \mathbb{R}^d$ whose changes have covariance matrix Σ. The $2d$ order types (one per asset and side) form a multivariate Hawkes process with

$$\lambda_t^j = \mu_j + \sum_{l=1}^{2d} K_{jl} \int_0^t \beta_j e^{-\beta_j (t-s)} \, \mathrm{d}N_s^l, \qquad j = 1, \dots, 2d, \tag{5}$$

where the off-diagonal blocks of K carry the excitation across assets and stationarity requires $\rho(K) < 1$. An order of type j raises the intensity vector to $\lambda^{+j} = \lambda + DKe_j$, with $D = \mathrm{diag}(\beta_1, \dots, \beta_{2d})$. An order of type $j = (i, s)$ fills the quote of asset i on side s with probability $f^{i,s}(\delta^{i,s})$. The market maker maximizes the risk-adjusted objective of the 2021 paper,

$$\mathbb{E}\Big[X_T + q_T^\top S_T - \frac{\gamma}{2} \int_0^T q_t^\top \Sigma \, q_t \, \mathrm{d}t\Big], \tag{6}$$

and the utility objective will be treated as a variant. With $\theta(t, q, \lambda)$ the value function after the cash and the marked-to-market inventory are factored out, the exact Hamilton-Jacobi-Bellman equation reads

$$0 = \partial_t \theta - \frac{\gamma}{2} q^\top \Sigma q - \sum_j \beta_j (\lambda_j - \mu_j) \, \partial_{\lambda_j} \theta + \sum_{j=(i,s)} \lambda_j \Big[\theta(t, q, \lambda^{+j}) - \theta(t, q, \lambda) + z_i H^{i,s}\big(p^j\big)\Big], \tag{7}$$

where $H^{i,s}(p) = \sup_\delta f^{i,s}(\delta)(\delta - p)$ and z_i is the trade size. The shadow price $p^j = [\theta(t, q, \lambda^{+j}) - \theta(t, q \pm z_i e_i, \lambda^{+j})]/z_i$ is taken at the post-order intensities, with the sign $+$ on the bid side and $-$ on the ask side. The forecast is again explicit,

$$m_t(u) = \bar{\lambda} + e^{-D(I-K)u}\big(\lambda_t - \bar{\lambda}\big), \qquad \bar{\lambda} = (I - K)^{-1}\mu, \tag{8}$$

so a burst of orders in one asset raises the forecast of the other assets through the off-diagonal blocks of K. Freezing the intensities at this forecast and applying the quadratic approximation of 2021 (its Proposition 1) gives the matrix Volterra-Riccati system

$$A'(\tau) = 2A(\tau)\Gamma_t(\tau)A(\tau) - \frac{\gamma}{2}\Sigma, \qquad A(T) = 0, \tag{9}$$

with the diagonal matrix

$$\Gamma_t(\tau) = \mathrm{diag}_i\Big(z_i\big[m_t^{i,b}(\tau - t)\, h_2^{i,b} + m_t^{i,a}(\tau - t)\, h_2^{i,a}\big]\Big), \qquad h_2^{i,s} = (H^{i,s})''(0), \tag{10}$$

so that $H^{i,s}(p) \approx h_0^{i,s} + h_1^{i,s} p + \frac{1}{2} h_2^{i,s} p^2$ as in the 2021 paper. The equations for B and C follow from those of the 2021 paper under the same substitution. The linear term B is driven by the difference between the bid and ask forecasts. After a burst of sell orders in one asset, the forecast of sell orders in the other assets rises through the off-diagonal blocks of K. The off-diagonal terms of A then pass the imbalance of one asset to the quotes of correlated assets. The quotes of the other assets are therefore skewed even before any order arrives in them.

With constant intensities equal to $\bar{\lambda}$, the long-horizon limit $A = \frac{1}{2}\Gamma^{-1/2}\left(\gamma\Gamma^{1/2}\Sigma\Gamma^{1/2}\right)^{1/2}\Gamma^{-1/2}$ recovers the closed form of 2021 (its Proposition 3), which serves as a check of every implementation. The quotes are computed from the exact optimizer of each Hamiltonian at the approximate shadow price, as in Barzykin (2026b). A second version replaces the quadratic Hamiltonian by the exact one, which removes the expansion error while keeping the frozen forecast. Its value function is no longer quadratic, so it is computed on the inventory grid, for two assets only, and tabulated over the current intensities so that it can be evaluated at every order of a Monte Carlo run. The comparison between the two versions answers the third detailed research question.

For two assets the exact equation has two inventory dimensions and four intensity dimensions, which is too large for a fine grid. The exact reference will therefore be built in two steps. First, reduced models keep the lifted state at three or four dimensions besides time, where the semi-Lagrangian scheme of Stage 1 applies. One reduced model has a single market-wide activity factor. The other has a sell-side and a buy-side memory variable shared by both assets, so that a burst of sell orders in one asset raises the sell intensity of the other; it carries the directional cross-asset excitation that Stage 2 is about. Second, for the full model with four memory variables, the study will try the generalised policy iteration of Cheridito, Dupret and Hainaut (2025) in its version with a continuous-time Bellman update. This version needs one sampled jump per point and was built for control problems with jumps, whereas the residual version becomes inefficient with jumps; residual-based solvers also failed on jump operators in the literature (Jain, Firoozye, Kochems and Treleaven, 2025) and drifted away from a good start in preliminary tests. The method works on continuous states, so it must be adapted to integer inventories. It will serve as a near-exact reference only where it matches the grid on both reduced models. If it fails this test, the full model, the models with more than two assets and the fitted models with a sum of two exponentials will be scored against the best policy found.

Reinforcement learning: Two kinds of agents will be trained in both stages. The first is the model-based actor-critic of Guéant and Manziuk (2019). Its state is the inventory and the intensities, and its critic is started from the closed-form value function, as the 2021 paper suggests. The second is

a model-free agent trained by proximal policy optimization with Hawkes-structured input features (Acero, Plank and Zhang, 2026), which does not know the kernel. Both are trained on an exact event simulator of the Hawkes flow, because preliminary experiments showed that an agent trained on a discretized version of the model inherits its errors. Every configuration will use at least five seeds fixed in advance, and every agent will be scored by the evaluation protocol below, never by its own critic.

Evaluation: All policies will be scored by Monte Carlo simulation of the true Hawkes model. Because every order excites the flow whether it fills the market maker or not, the order stream does not depend on the policy. It is simulated once, together with the price path, and each order carries a uniform random number that decides whether it fills a quote at a given distance. All policies then face the same orders, the same prices and the same numbers, and differences between policies are measured with paired standard errors far smaller than the values themselves. The main measure is the regret defined in Section 1.2: against the exact solution for one asset and for the reduced two-asset models, and against the best policy found in the other cases. The inventory distribution, the average spread and skew, and the computing time per quote will also be reported. The parameters will be varied along five directions: the speed of the memory (through β, with the mean arrival rate and the branching ratio held fixed), the branching ratio or spectral radius $\rho(K)$, the share of cross-asset excitation, the correlation of the prices, and the risk aversion.

Verification: Every derivation will be re-derived independently and every closed form will be checked in two ways: it must reduce to the 2013 or 2021 result when $K = 0$, to machine precision, and it must agree with an independent ordinary differential equation solver. Each exact solver will be checked against a second solver or simulator. Where the problems actually solved differ from the original papers, for example in the horizon or in the objective, this will be stated plainly.

Scope: The study keeps the convention that all orders excite the flow. The case where only the market maker's own fills excite it, in which the quadratic ansatz does not close (Bergault, Bertucci, Bouba, Guéant and Guilbert, 2024), will be treated only by the exact and learning methods if time allows. Queue position in the order book and power-law kernels, beyond sums of a few exponentials, are outside the scope.

4.2 Data sources

The fitted part of the study needs order flow data in which every market order carries a time stamp and the side of the aggressor, for at least two related assets traded at the same time. It also needs best quotes or order book data, to measure how the fill probability falls with the distance of a quote. The

datasets below were selected against these needs after a search of public archives and commercial providers. Coverage and prices were checked on the providers' own pages in October 2026 and will be confirmed when the data are ordered.

Simulated data: Order flow will be simulated exactly by thinning (Ogata, 1981), with the exact exponential decay of the intensities between events, for every parameter set of the study.

Accessible datasets: Table 2 lists the free sources. The Binance archive is the main free source; the others serve as checks across venues and as pilots of the paid datasets.

Dataset	Content and coverage	Access	Use in this study
Binance futures public archive (data.binance.vision)	Every trade with a millisecond time stamp and a flag for the aggressor side, from 31 December 2019 for BTCUSDT and ETHUSDT; best bid and ask updates from 16 May 2023 to 30 March 2024; aggregate depth within 1% to 5% of the mid, about once a minute, from 1 January 2023	Free daily and monthly files with checksums	Main free source: joint fit of the two-asset flow; fill decay from the period with best quotes
Bybit and OKX public trade data	Trades with the aggressor side; Bybit as daily files, OKX through its trade history interface; no historical order book	Free	Check that cross-asset excitation is not specific to one venue
Deribit history interface	Every trade of its crypto futures and options since the start of the platform	Free	Optional check across futures and options
Hyperliquid node data	Trade fills from March 2025 and monthly order book snapshots in public cloud storage	Free; the requester pays the transfer cost	Check on a decentralized venue
Tardis.dev samples and trial	Trades with the taker side and order book updates with microsecond time stamps for more than 50 venues; the first day of each month without a key; a one-month trial covering days 1 to 10 of the previous month	Free	Pilot of the full pipeline before purchase
Databento sign-up credit	A credit of $125, valid six months; by the provider's figures it covers about 14 months of order-by-order data for one Chicago Mercantile Exchange product, such as the E-mini S&P 500 future	Free credit	Pilot of the equity index futures pair
Nasdaq TotalView-ITCH sample files	Full message files for sample trading days	Free	Test of the equity order book pipeline

Table 2: Accessible datasets.

Datasets to be purchased: Table 3 lists the commercial and licensed sources. The first two are part of the plan; the last two are extensions to equities and to the over-the-counter bond market, the setting closest to the 2021 paper.

Dataset	Content and coverage	Cost and terms	Use in this study
Tardis.dev, academic plan	Tick-level trades with the taker side, order book updates and snapshots, and best quotes for more than 50 venues, with Binance futures from 17 November 2019	About $350 per month for perpetual contracts, billed quarterly or yearly; quarterly billing gives 12 months of history; files only, no single-day purchases	Main paid source: full order books for the fill decay, the BTCUSDT and ETHUSDT pair across venues
Databento, Chicago Mercantile Exchange order-by-order data	Every order event with the aggressor side and nanosecond time stamps; also trades and best quotes; Eurex and Nasdaq data from the same provider	Usage-based, from $0.50 per gigabyte; monthly plan from $199	Second asset class: E-mini S&P 500 and E-mini Nasdaq-100, and 10-year and 30-year Treasury note and bond futures
LOBSTER (Nasdaq order books)	Reconstructed order books and full message files for every Nasdaq stock, from 27 June 2007	For academic research only; price on request	Extension to equity pairs from one sector
Trade Reporting and Compliance Engine (Financial Industry Regulatory Authority)	Every corporate bond trade with the buy or sell side and a dealer or customer flag; the academic product has a 36-month delay	Academic product: $500 per year plus a $500 set-up fee at its launch in 2017; also through Wharton Research Data Services	Extension to dealer-to-client flow, with excitation across bonds of one issuer

Table 3: Datasets to be purchased.

Alternatives and datasets not used: Other providers offer comparable data with prices on request or through institutional licences: the Chicago Mercantile Exchange DataMine service, the Deutsche Börse A7 platform and Data Shop for Eurex and Xetra order-by-order data, LSEG Tick History, the Daily Trade and Quote files of the New York Stock Exchange through Wharton Research Data Services, and CoinAPI and Kaiko for crypto order books. They will be used only if one of the planned sources fails. Some well-known datasets do not fit this study. The FI-2010 benchmark has normalized order book snapshots of five Finnish stocks over ten days, without trade sides. Order book snapshots aggregated at one second, such as those shared on Kaggle, lose the event times. Foreign exchange tick data from Dukascopy and TrueFX contain quotes but no trades. Request-for-quote data, the flow of the 2021 paper and of Barzykin (2026b), are proprietary to dealers and platforms.

Fitting: On the market data, a four-variate Hawkes model (two assets, two sides) with exponential kernels, and a version with a sum of two exponentials, will be fitted by maximum likelihood on

order bursts in 100 millisecond bins, over several trading days rather than short windows. Bursts of the two assets that fall in the same bin are ties, which an exponential kernel cannot attribute to excitation, and their treatment drives the estimate of the cross-asset block of K. The fit will therefore be repeated with random ordering within the bin and with finer bins, and compared with a version that adds a common shock hitting both assets at once, in the spirit of the synchronising factor of Choi, Jang, Lee and Zheng (2021). The fit will be tested with time-rescaled inter-event times and with variance-to-mean ratios of order counts. The volatilities, the price correlation and the fill decay will be estimated from the same sources.

Benchmark parameter sets from the literature: the reference case of the 2013 paper ($\sigma = 0.3$ tick per square-root second, $\Lambda = 0.9$ per second, $k = 0.3$ per tick, $\gamma = 0.01$ per tick, $T = 600$ seconds), with the Poisson flow replaced by a Hawkes flow of the same mean rate; the price and request parameters of the two-asset example of the 2021 paper (Section 6), simplified to one request size per asset, no drift and a reduced inventory grid, with its Poisson requests replaced in the same way; and the three exponential regimes of Barzykin (2026b), whose exact values were reproduced in the preliminary work.

4.3 Expected Results

- ➢ A forecast-driven version of the 2013 linear system for Hawkes order flow, with its exact counterpart and a map of its error.
- ➢ A matrix Volterra-Riccati approximation of the 2021 model with cross-asset excitation, in closed form up to one matrix Riccati equation per update.
- ➢ Exact solutions for one asset and for reduced two-asset models, and a near-exact solution of the full two-asset model if the neural solver passes its validation.
- ➢ A map of where each method (closed form, exact solution, reinforcement learning) stays close to the optimum, as a function of the speed of the memory, the branching ratio, the cross-excitation, the price correlation and the risk aversion.
- ➢ Fitted cross-asset Hawkes models for a pair of cryptocurrency futures and a pair of equity index futures, and the comparison of all policies on them.
- ➢ Open code and run records from which every reported number can be reproduced.

5. Project Timelines and Budget Plan

5.1 Research Timeline

Period	Activities
Week 1	The 2013 model with Hawkes order flow: forecast-driven linear system solved on the 2013 reference case, with the check $K = 0$. The 2021 model with cross-asset Hawkes flow: matrix Volterra-Riccati system solved on a two-asset case, with the check $K = 0$
Week 2	Data: free Binance files, Tardis.dev and Databento pilots, orders for the paid data; Hawkes fits including the cross-asset block; exact single-asset solution under the utility objective
Week 3	Single-asset Monte Carlo comparison of the closed forms and the exact solution; exact solutions of the reduced two-asset models
Week 4	Two-asset Monte Carlo comparison on the reduced models; independent verification of the derivations and solvers
Week 5	Neural solver for the full two-asset model and its validation on the reduced models (fallback: loss against the best policy found)
Weeks 6 to 7	Reinforcement learning agents in both stages, five seeds per configuration
Week 8	Parameter sweeps and the comparison on the fitted models, including the equity index futures pair
Weeks 9 to 10	Analysis, discussion and draft writing
Week 11	Internal review and revisions
Week 12	Submission and dissemination (working paper, seminar and code release)

5.2 Budget Plan

Task Category	Description	Estimated Budget
Crypto market data	Tardis.dev academic plan for perpetual contracts, one quarter	$1050
Futures market data	Databento order-by-order data beyond the free credit	$200
Equity market data	LOBSTER academic access for Nasdaq order books (price on request; estimate)	$350
Bond market data	Trade Reporting and Compliance Engine academic product, first year including set-up	$1000
Computing	Cloud processor time for the exact solvers and the Monte Carlo runs; graphics processor time for training the learning agents	$450
Data storage	Storage of several months of trade and order book files	$100
Literature access	Papers and theses not available in open access	$100
Conferences and publication	Registration for one conference in mathematical finance or in machine learning for finance	$500
Contingent expenses	For unconsidered research costs	$250
Total estimated budget		**$4000**

6. Project Deliverables and Dissemination plan

A short report will be prepared at the end of each stage. The main deliverable is a full research report containing the two models, their closed-form approximations, the exact and learning solutions,

the comparison on simulated and fitted order flow, and a clear statement of what is proved, what is verified numerically and what remains open. The report will be turned into a working paper for the arXiv quantitative finance section and submitted to a journal in mathematical finance, such as Applied Mathematical Finance, where the 2021 paper appeared, or Quantitative Finance. The results will also be presented at one conference in mathematical finance or in machine learning for finance. The code, the fitted model parameters and the run records will be released in a public repository, so that the exact solutions can serve as benchmarks for other researchers who test learning methods on Hawkes order flow. Purchased data will not be redistributed; only derived statistics and fitted parameters will be published. A seminar presentation with speaking notes will present the results.

7. Ethical Aspects and Research Integrity

7.1 Ethical Aspects

This research will follow the usual ethical standards for research that uses market data. The study uses anonymous market data (trade and quote records published by exchanges or sold by data providers) and simulated data. These data contain no personal information, so no privacy concerns arise. The strategies studied will be tested only in simulation and on historical data; no orders will be sent to a live market and no real money will be traded.

- Every dataset will be used under the terms of its provider; academic licences restrict use to research and forbid redistribution of the raw data.
- No personal or confidential data will be used at any stage.
- Data and results will be stored securely and used only for research purposes.

The study also commits to transparency by documenting its methods and releasing its code. No conflict of interest or financial bias is expected, and any institutional collaboration will be declared.

7.2 Research Integrity and Reproducibility

Every number in the reports will come from a recorded run, with its seed, configuration and code version. Results will be separated into what is proved, what is verified numerically, what is plausible and what is open. Single-seed results will not be used for conclusions. When the problem actually solved differs from the original one, for example in its horizon or its objective, this will be stated. Claims that a combination has not been studied will be phrased as not found by the search, since a search cannot exclude theses or unpublished drafts. Errors found after a report is delivered will be corrected openly.

7.3 Expected Ethical Outcomes

- Quoting rules for clustered order flow that are tested against exact solutions before any practical use.
- Published exact values that other researchers can use to check reinforcement learning results on Hawkes order flow.
- Reports in which every number can be traced to a recorded run.

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